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Kelas : IF – 1

Rangkuman praktikum system terdesentralisasi dan terdistribusi Simulasi Arsitektur Client-Server Menggunakan Docker di AWS EC2

```
Running kernel seems to be up-to-date.  
No services need to be restarted.  
No containers need to be restarted.  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-66-247:~$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 100.24.115.34 PrivateIPs: 172.31.66.247

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-66-247:~$ sudo systemctl start docker  
Warning: The unit file, source configuration file or drop-ins of docker.service changed on disk. Run 'systemctl daemon-reload' to reload units.  
ubuntu@ip-172-31-66-247:~$ sudo systemctl enable docker  
ubuntu@ip-172-31-66-247:~$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 100.24.115.34 PrivateIPs: 172.31.66.247

Pembahasan : Update sudo system sudo apt update sudo apt install docker.io git -y

```

ubuntu@ip-172-31-66-247:~$ sudo systemctl enable docker
ubuntu@ip-172-31-66-247:~$ sudo docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
ubuntu@ip-172-31-66-247:~$ git clone https://gitlab.com/akbarwijaya/aerostack.git
Cloning into 'aerostack'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 15 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 4.05 KiB | 1.35 MiB/s, done.
ubuntu@ip-172-31-66-247:~$ cd aerostack
ubuntu@ip-172-31-66-247:~/aerostack$ sudo docker build -t web-app-ku .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  82.43kB

```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 100.24.115.34 PrivateIPs: 172.31.66.247

```

Run 'docker run --help' for more information
ubuntu@ip-172-31-66-247:~/aerostack$ sudo docker build -t web-app-ku .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  82.43kB
Step 1/9 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/9 : WORKDIR /app
--> Using cache
--> 82cbd31cbfb1
Step 3/9 : ENV PYTHONDONTWRITEBYTECODE=1
--> Using cache
--> 97fa42a51f11
Step 4/9 : ENV PYTHONUNBUFFERED=1
--> Using cache
--> f15443f4b159
Step 5/9 : COPY requirements.txt .
--> Using cache
--> 7b9b16a072e9
Step 6/9 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> d40bec1e3365
Step 7/9 : COPY . .
--> Using cache
--> 3563104228a4
Step 8/9 : EXPOSE 80
--> Using cache
--> dacee4bb12a7
Step 9/9 : CMD ["python", "app.py"]
--> Using cache
--> 0f6b3658b2c1
Successfully built 0f6b3658b2c1
Successfully tagged web-app-ku:latest
ubuntu@ip-172-31-66-247:~/aerostack$

```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 100.24.115.34 PrivateIPs: 172.31.66.247

```
ubuntu@ip-172-31-66-247:~/aerostack$ sudo docker run -d -p 80:80 web-app-ku
c96b7161b2b8e6642b3d26d0f6ff4a28019cf820b9edaa34d5bb9d8e9da6b90f
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint eloquent_neumann (89a152d16b345c7f9731c9de75a119704e55480df1a1f3e0ca452156d8e9f2f5): failed to bind host port for 0.0.0.0:80:172.17.0.2:80/tcp: address already in use

Run 'docker run --help' for more information
ubuntu@ip-172-31-66-247:~/aerostack$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 100.24.115.34 PrivateIPs: 172.31.66.247

Pembahasan : Menjalankan service docker sudo systemctl start docker

sudo systemctl enable docker sudo docker --version

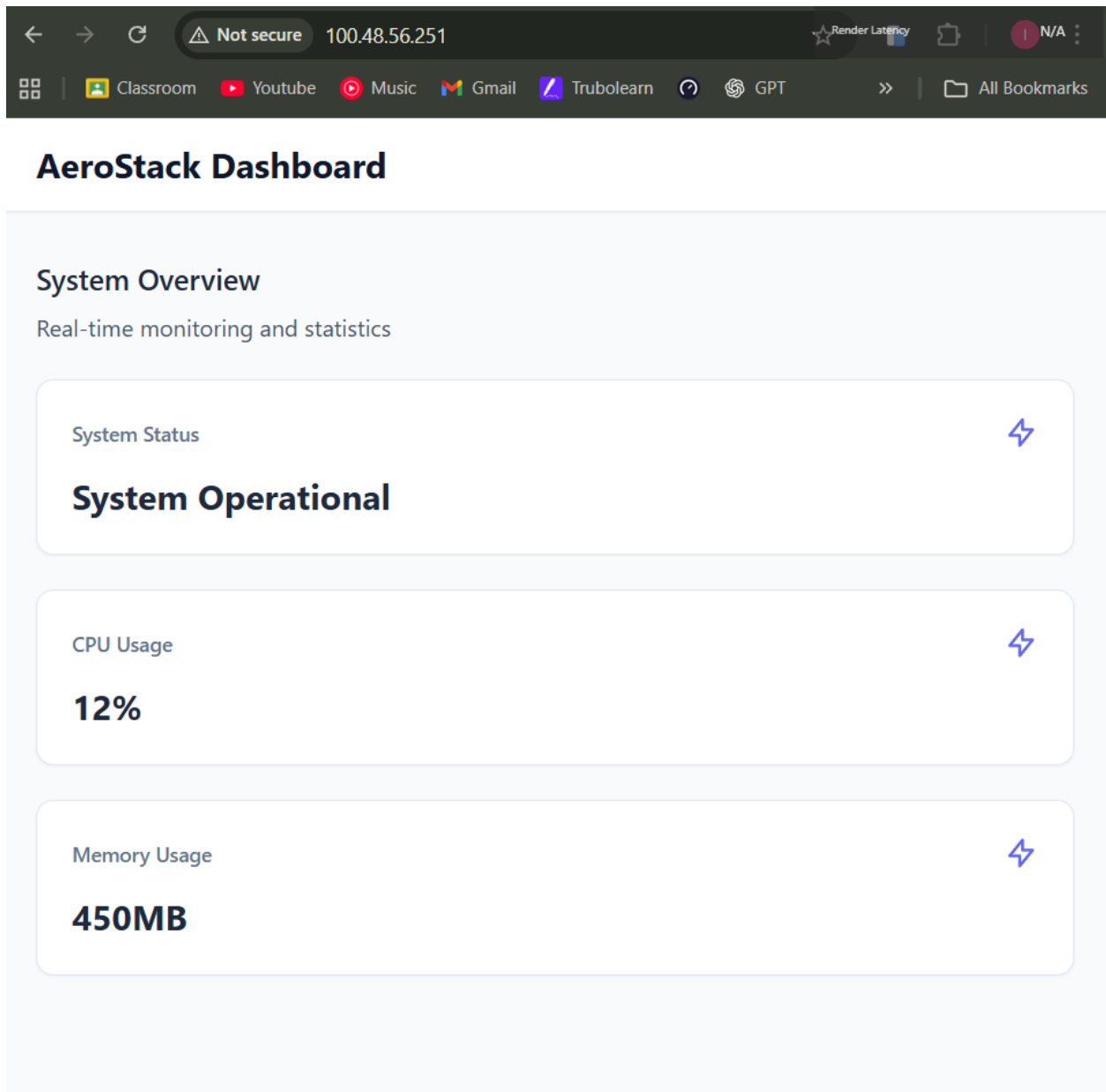
```
Last login: Thu Dec 11 08:41:30 2025 from 18.206.107.27
ubuntu@ip-172-31-66-247:~$ cd aerostack/
ubuntu@ip-172-31-66-247:~/aerostack$ sudo docker run -d -p 80:80 web-app-ku
3dc491ae074f4729892f3a78751cdca24aae82e517c44dlfc45e8562cb4d3ff1
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint awesome_zhukovsky (bfb4b26a677081c756244ccaaa080fa9d45e87f31829d4a8beb9eb7f76986a): failed to bind host port for 0.0.0.0:80:172.17.0.2:80/tcp: address already in use

Run 'docker run --help' for more information
ubuntu@ip-172-31-66-247:~/aerostack$ sudo lsof -i :80
COMMAND PID    USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
apache2  664    root    4u    IPv6  6489      0t0  TCP *:http (LISTEN)
apache2  713  www-data  4u    IPv6  6489      0t0  TCP *:http (LISTEN)
apache2  715  www-data  4u    IPv6  6489      0t0  TCP *:http (LISTEN)
apache2  716  www-data  4u    IPv6  6489      0t0  TCP *:http (LISTEN)
apache2  720  www-data  4u    IPv6  6489      0t0  TCP *:http (LISTEN)
apache2  721  www-data  4u    IPv6  6489      0t0  TCP *:http (LISTEN)
ubuntu@ip-172-31-66-247:~/aerostack$ sudo systemctl stop apache2
sudo systemctl disable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-instal
tall.
Executing: /usr/lib/systemd/systemd-sysv-install disable apache2
Removed "/etc/systemd/system/multi-user.target.wants/apache2.service".
ubuntu@ip-172-31-66-247:~/aerostack$ sudo systemctl stop apache2
sudo systemctl disable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-instal
tall.
Executing: /usr/lib/systemd/systemd-sysv-install disable apache2
ubuntu@ip-172-31-66-247:~/aerostack$
```

i-0a454cf2bfe79bea1 (vm-server-application)

PublicIPs: 100.48.56.251 PrivateIPs: 172.31.66.247

Pembahasan : Troubelshoting sedikit karena port 80 terpakai xampp



Pembahasan : Membuka di web dengan ip public server

```
Fetched 13.3 MB in 2s (5388 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-24-75:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.5.0-2ubuntu10.6).
0 upgraded, 0 newly installed, 0 to remove and 30 not upgraded.
```

i-000a8ff1c561f54d3 (Vm-client)

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75

```
curl is already the newest version (8.5.0-2ubuntu10.6).
0 upgraded, 0 newly installed, 0 to remove and 30 not upgraded.
ubuntu@ip-172-31-24-75:~$ curl http://172.31.66.247/api/status
{
  "data": {
    "cpu": "12%",
    "memory": "450MB"
  },
  "message": "System Operational",
  "status": "success"
}
ubuntu@ip-172-31-24-75:~$
```

i-000a8ff1c561f54d3 (Vm-client)

PublicIPs: 44.210.41.98 PrivateIPs: 172.31.24.75

Pembahasan : tes install curl di client dan memanggil dari client menggunakan curl dan private IP dari server berhasil

Kesimpulan : mensimulasikan arsitektur client-server menggunakan Docker di lingkungan AWS EC2. Melalui konfigurasi dua virtual machine yang terpisah, yaitu VM1 sebagai server aplikasi dan VM2 sebagai client testing, praktikum ini mendemonstrasikan bagaimana membangun dan mengelola infrastruktur berbasis kontainer dalam skenario jaringan cloud. Implementasi Docker pada VM1 memungkinkan deployment aplikasi web (AeroStack) yang efisien dan terisolasi, sementara konfigurasi security group yang tepat pada AWS EC2 memastikan aksesibilitas dan keamanan sistem. Pengujian konektivitas antara VM2 dan VM1 menggunakan private IP menunjukkan komunikasi internal yang berhasil dalam jaringan AWS,

